

CATARMAN, Philippines

WMO: 985460

Lat: 12.5056N

Lon: 124.6285E

Elev: 6

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.5	73.7	69.6	108.6	75.2	70.7	113.1	75.8	22.8	76.4	18.9	77.4	2.7	180	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.7	92.7	81.9	91.6	81.8	90.6	81.7	83.5	89.7	82.9	89.0	82.6	88.7	6.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.8	165.2	87.6	81.2	162.1	87.2	80.8	160.0	86.9	47.5	90.2	46.9	89.5	46.4	89.2	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.2	14.2	12.3	DB	70.5	94.8	1.9	1.0	69.1	95.5	68.0	96.1	66.9	96.7	65.5	97.4	(4)
			WB	68.7	84.5	2.4	0.9	66.9	85.1	65.5	85.6	64.2	86.1	62.5	86.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	79.2	79.5	80.3	81.8	83.0	83.0	82.5	83.1	82.6	81.8	81.1	80.0
	DBStd	2.20	1.56	1.69	1.59	1.52	1.54	1.92	1.95	1.95	1.86	1.70	1.62	1.69
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11502	904	827	939	955	1024	989	1007	1027	979	987	933	931
	CDD65	6027	439	407	474	505	559	539	542	562	529	522	483	466
	CDH74	63364	3677	3629	4633	5616	6647	6156	5988	6391	5846	5578	4901	4302
	CDH80	20467	558	756	1267	1961	2684	2407	2204	2475	2142	1806	1365	842
Wind	WSAvg	5.0	7.5	6.5	5.7	4.6	3.8	3.5	3.8	4.2	3.8	4.3	5.1	7.0
Precipitation	PrecAvg	140.5	18.4	11.7	11.3	6.4	6.3	8.7	8.9	6.8	8.8	12.5	17.5	23.2
	PrecMax	265.9	55.9	46.9	51.3	23.9	30.9	18.0	17.6	15.0	19.4	22.8	39.4	40.0
	PrecMin	78.0	4.6	1.5	1.5	0.6	1.0	3.4	2.1	1.7	2.0	5.8	4.0	4.6
	PrecStd	43.4	10.4	11.2	10.8	5.7	5.9	4.1	4.2	3.1	4.1	4.9	6.3	9.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	88.3	89.6	90.7	93.3	94.1	93.0	94.1	93.1	91.6	89.3	87.7
		MCWB	80.3	80.6	82.4	82.3	82.3	81.9	82.0	81.4	81.9	81.7	81.3	81.5
	2%	DB	85.3	86.6	87.9	89.6	91.7	92.2	91.3	91.9	91.3	89.8	88.1	86.3
		MCWB	79.7	80.0	81.0	81.6	82.2	82.0	81.7	81.4	81.6	81.5	81.2	80.8
	5%	DB	84.3	85.4	86.9	88.7	90.5	90.8	89.9	90.7	90.0	88.6	87.1	85.4
		MCWB	79.1	79.4	80.2	81.3	82.0	81.9	81.4	81.3	81.3	81.2	81.0	80.3
	10%	DB	83.1	84.3	85.8	87.7	89.3	89.3	88.6	89.3	88.7	87.3	86.0	84.2
		MCWB	78.6	78.7	79.5	81.0	81.7	81.7	81.0	81.0	80.9	80.8	80.6	79.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	81.9	83.5	83.6	83.7	83.7	83.5	83.5	83.5	83.3	83.0	82.7
		MCDB	85.1	86.2	88.5	89.1	90.2	90.5	90.4	90.5	90.3	88.9	87.1	85.9
	2%	WB	80.8	81.0	81.9	82.7	83.2	83.0	82.6	82.7	82.6	82.4	82.1	81.7
		MCDB	84.1	85.2	86.7	88.2	89.7	89.8	89.4	89.6	89.3	88.0	86.5	85.2
	5%	WB	80.0	80.2	80.9	81.9	82.6	82.4	81.9	82.0	81.9	81.7	81.5	80.9
		MCDB	83.1	84.2	85.7	87.4	89.1	89.3	88.4	88.7	88.4	87.1	85.9	84.3
	10%	WB	79.1	79.3	80.1	81.3	82.0	81.8	81.3	81.4	81.2	81.0	80.9	80.1
		MCDB	82.1	83.0	84.7	86.7	88.3	88.3	87.5	87.9	87.4	86.3	85.1	83.3
Mean Daily Temperature Range	5% DB	MDBR	7.2	9.1	10.6	11.7	12.7	12.4	11.6	11.9	11.8	10.6	9.6	7.6
		MCDBR	9.5	11.4	12.9	13.2	14.6	14.6	13.9	14.1	13.9	12.8	11.6	9.8
		MCWBR	5.6	6.7	7.7	7.3	7.5	7.0	6.7	6.3	6.5	6.6	6.5	5.9
	5% WB	MCDBR	8.6	10.4	12.0	12.5	13.5	13.6	13.0	12.9	12.8	11.8	11.0	9.3
MCWBR		5.9	6.7	7.7	7.5	7.5	7.2	7.0	6.8	6.9	6.7	6.8	6.2	
Clear-Sky Solar Irradiance	taub	0.466	0.442	0.440	0.429	0.429	0.427	0.432	0.458	0.448	0.454	0.450	0.469	
	taud	2.273	2.355	2.375	2.410	2.438	2.455	2.440	2.351	2.379	2.354	2.357	2.268	
	Ebn at Noon	265	277	279	279	275	272	272	268	273	270	268	260	
	Edh at Noon	42	40	40	39	37	36	36	40	39	40	39	42	
All-Sky Solar Radiation	RadAvg	1216	1521	1715	1927	1853	1675	1533	1625	1563	1446	1350	1173	
	RadStd	169	196	204	140	150	96	130	92	144	99	124	119	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page